Control Desing
Spring 2026

Assessment Task 5

Final Project Demonstration

A/Prof Ricardo P. Aguilera

Assessment Criteria

- Students are required to present their design and implementation of two control systems:
  - Frequency-domain based design (using transfer functions)
  - Time-domain based design (using SFC and observer)

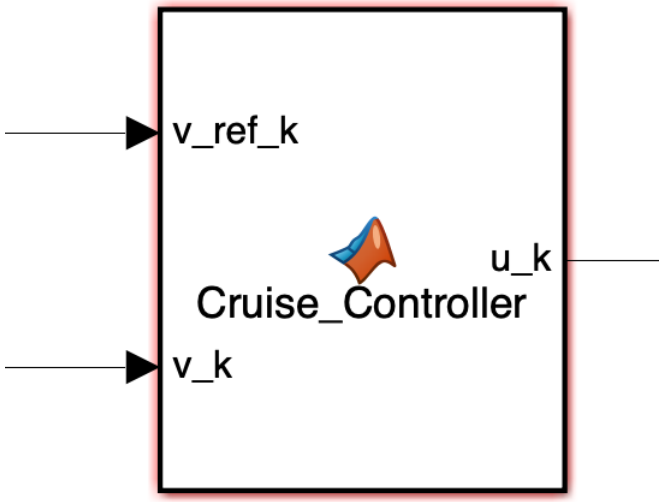
  Each control system must be coded in a single MATLAB Function block (not using blocks)

- Total final demo time for each team is 15mins. This is divided in three main parts:
  - 1) Presentation (5mins),
  - 2) Control System Demonstration (5mins), and
  - 3) Q&A (5mins).

Assessment Criteria

All control systems must be coded in a single MATLAB function.

If you have reference filters, controllers, etc., all these stages must be inside a single MATLAB Code.




```
1   function u_k = Cruise_Controller(v_ref_k, v_k, Ka, Kb,kf1, kf2)
2   %% Initialization
3   persistent e_k_1 u_k_1 v_ref_f_k_1
4   if isempty(e_k_1)
5       e_k_1 = 0;  u_k_1 = 0;
6       v_ref_f_k_1 = 0; % Previous value of the speed reference
7   end
8   %                       v_ref_f_k +             u_k
9   % v_ref_k -->[F(z)]-------------->O-->[C(z)]----->[Go(z)]----> v_k
10  %                                  -|                       |
11  %                                   ----------------------
12  %% F(z) reference filter,
13  % v_ref_f_k:   filtered version of the reference in instant k
14  % v_ref_f_k_1: filtered version of the reference in instant k-1
15
16  v_ref_f_k = kf1*v_ref_k + kf2*v_ref_f_k_1;
17
18  %% C(z): PI Controller
19  e_k = v_ref_f_k - v_k; % present error
20  u_k= Ka*e_k+Kb*e_k_1 + u_k_1;
21
22  %% Memory
23  e_k_1 = e_k;
24  u_k_1 = u_k;
25  v_ref_f_k_1 = v_ref_f_k;
```

Assessment Criteria

1) Presentation (5mins):

- Students must explain their system modelling, control systems designs, simulations, and implementations. This time must be equally spread among team members.
- During the presentation, students are free to choose any supporting tool, e.g., equations, figures, plots, videos, origamis, puppets, etc.
- Please, do not use the presentation time to explain control concepts. The presentation must be focused on your control system designs.
- During the presentation, please avoid reading from a script. Instead, explain your design in your own words and speak spontaneously.

Assessment Criteria

2) Control Systems Demonstration

- Students are required to demonstrate their two control system designs. This demonstration considers MATLAB simulation and practical implementation.
- All MATLAB/Simulink/Arduino files must be uploaded to a folder in MS Teams. Include individual subfolders for each controller.
- Since this is an individual assessment, each student must be able to:
  - run each MATLAB simulation and/or Arduino code from their own computer without requiring help from any other team member.
  - change the controller settings in MATLAB and/or Arduino IDE as requested by the Teaching Team.

Assessment Criteria

2) Questions

- After demonstrations, students are required to answer technical questions from the teaching team. Since this is an individual assessment, each team member must be able to attend questions/comments related to:
  - explain how the nonlinear/linear model of the system to be controlled was obtained.
  - explain, using control engineering concepts, how each of the chosen controllers have been selected (type of transfer function, state-space model) and designed to achieve a desired control target (not trial and error, not MATLAB autotuning).
  - answer, using control engineering concepts, any question or concern risen by the teaching team related to your project.

Assessment Criteria

- Some typical questions you may be asked:
  1. How did you select the controller sampling time?
  2. Is your system open-loop stable or unstable ?
  3. How do you reduce a closed-loop overshoot without changing the closed-loop poles?
  4. How do you select the observe poles?
  5. How do you translate continuous-time eigen values into discrete-time ones?
  6. Can you still daydream at night?
  7. If we expect the unexpected, doesn't the unexpected become expected?
  8. If I try to fail, but I succeed, which one did I actually do?

  Please, ensure all MATLAB and Arduino files run properly before the demonstration.

  All the best!!!